

Executive MBA

Financial Accounting

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Lecture 6: Financial Reporting for Lease Contracts



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What is off-balance sheet financing?

- An method whereby a company obtains financing but does not recognize a liability on its balance sheet.
- Why do some firms want to avoid recognizing debt on the balance sheet?
 - Reduce perceived riskiness of the firm to increased access to new debt and lower the costs of borrowing
 - Maintain slack in existing debt covenants
- **Goal of GAAP:** Economic substance over legal form
 - But far from successful all the time

Leases

- Consider the following transactions:
 1. Lufthansa purchases an aircraft with an expected useful life of 40 years from Airbus.
 2. Lufthansa signs a non-cancelable thirty-five year contract with Airbus for the *right of use* of an aircraft with an expected useful life of 40 years.
- Is there a distinction between these two transactions?
 - They differ in ***form***: transaction (1) transfers title of aircraft from Airbus to Lufthansa while for transaction (2), there is no title transfer.
 - However, both transactions have ***similar economic substance***: for both transactions, it is Lufthansa and ***not*** Airbus that enjoys substantially all the benefits and bears substantially of the economic risks of ownership

Leases

- A lease is a contractual agreement between a lessor and a lessee, that gives the **lessee** the right to use specific property, owned by the **lessor**, for a specified period of time in exchange for consideration.
 - A lease contract is an example of a *mutually unperformed contract*.
- Common Lease Provisions:
 1. Lessee has unrestricted right to use the asset during the lease term.
 2. Lessee agrees to make periodic payments to the lessor and to maintain the asset.
 3. Lessor retains title to the leased asset.

Why Lease vs. Buy?

Advantages of Leasing: Lessee

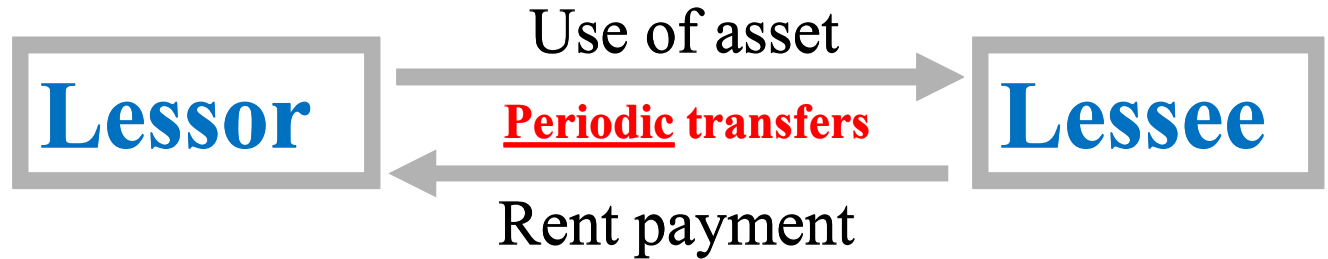
- 100% financing at fixed rates.
- Protection against obsolescence.
- Flexibility in terms of payments.
- Less costly financing.

Advantages of Leasing: Lessor

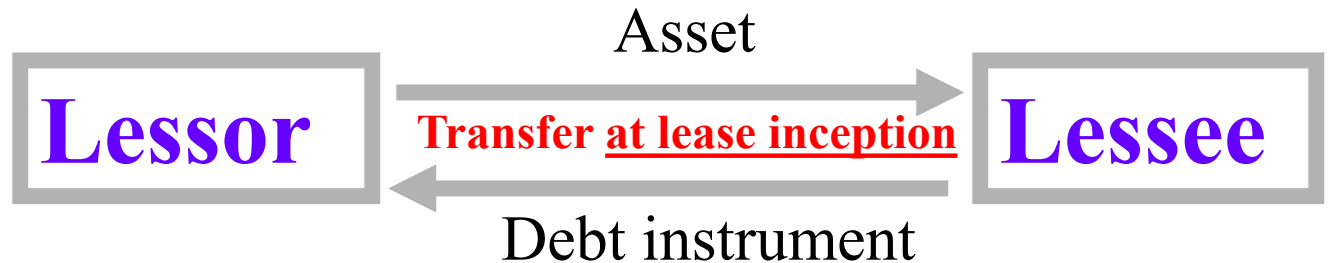
- Often provides profitable interest margins.
- It can stimulate sales of a lessor's product.
- It often provides tax benefits to various parties in the lease.
- It can provide a high residual value to the lessor.

Two views of leases...

No Capitalization:
Ongoing
operating cost/
executory
contract



Capitalization:
In-substance
asset purchase



Financial Reporting for Leases

- The key financial reporting issue for leases is to determine which party to the lease controls the asset, i.e., which party enjoys *substantially all of the benefits and the economic risks of ownership* in the leasing contract?
 - If the lessor (owner of the asset) controls the asset, then the lease contract is treated as a rental contract. Otherwise, it is treated as an installment asset purchase (sale) from the perspective of the lessee (lessor).
- Former Leases Standard (i.e., before fiscal 2019) specified *bright line* criteria to determine which party controlled the asset.
 - Most lessees actively structured the contracts around GAAP to avoid recognizing lease contracts as in-substance purchases/sales. Consequently, *under former leases standards, most lease contracts were kept off-balance sheet and therefore treated as operating leases (i.e., as long-term rental contracts)* even though they were in-substance purchases by the lessees.

How did Standard and Poor's (S&P) view Operating Leases under the Former Leases Standard?

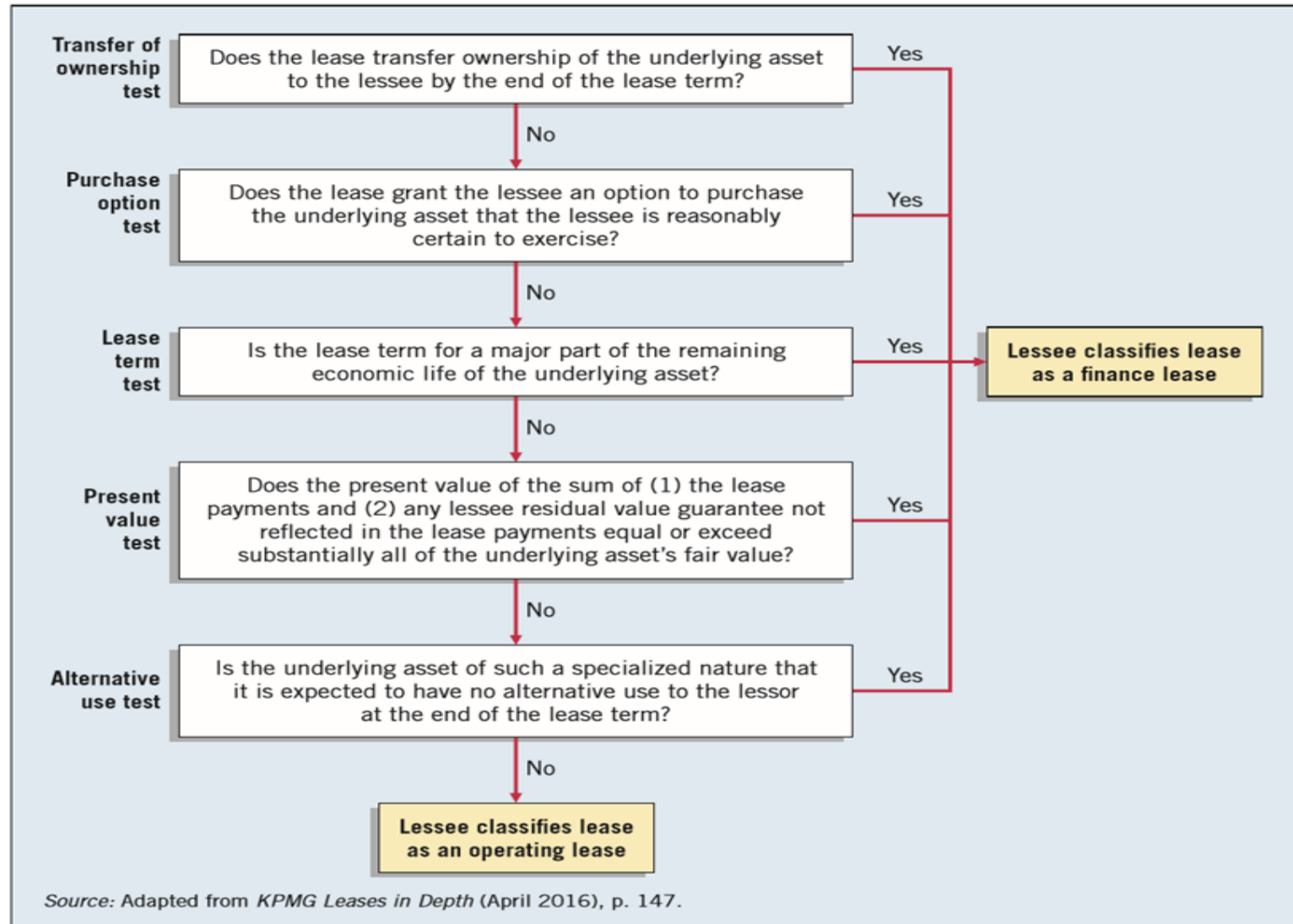
- “We view the accounting distinction between operating and capital leases as substantially artificial. In both cases, the lessee contracts for the use of an asset, entering into a debt-like obligation to make periodic rental payments...”
- Our operating lease adjustment therefore seeks to capitalize all leases, thereby:
 - enhancing comparability of operating and financial results and financial obligations among companies that lease assets under leases accounted for as operating or financing leases, or use debt to finance asset acquisition.
 - bringing financial ratios closer to the underlying economics by taking into consideration all financial obligations incurred, on or off the balance sheet.
 - improving analysis of how profitably a company employs its leased/owned assets.

Source: S&P's Analytical Approach to Leases, 2006

Current Leases Standard: Effective Fiscal 2019

- All contracts for less than or equal to 12 months would be treated as *short-term leases*, and not capitalized, consistent with the general use of the former category “operating lease” that was treated as a rental contract.
- All lease contracts > 12 months would be treated as *long-term leases and therefore capitalized* on balance sheets as Right-of-Use assets and Lease Liabilities. However...
 - IFRS: All long-term leases would be treated as Finance leases.
 - US GAAP: All long-term leases would be classified as either a finance lease or an operating lease:
 - **Finance leases** transfer **control of the lease asset** to the lessee. Finance leases are effectively like purchasing the asset and financing the purchase with a collateralized loan.
 - **Operating leases** transfer **control of the use of the lease asset**, but not the asset itself. Any lease of more than 12 months *not classified* as a finance lease is classified as an operating lease.

US GAAP: Finance vs Operating Leases



Leases Standard: Lease Accounting

- Under US GAAP, *regardless of whether lessee treats lease contract as finance or operating*, lessee would capitalize the lease contract at inception by recognizing a right-of-use asset and a lease obligation at inception of the lease.
- However, the income statement treatment of an operating vs. a finance lease is different.
 - For a *Finance Lease*: lessee/lessor accounting is similar to that of an installment purchase/sale:
 - Lessee recognizes interest expense using the effective interest method and amortization expense on the leased asset while Lessor typically recognizes profit/loss when asset is derecognized and interest revenue over time.
 - For an *Operating Lease*: lessee also measures interest expense using the effective-interest method. However, the lessee amortizes the right-of-use asset such that the total lease expense is the **same from period to period**.
 - Consequently, only a single lease expense (comprised of interest on the liability and amortization of the right-of-use asset) is recognized on the income statement.
 - However, lessor accounting is similar to that of a long term rental contract: lessor doesn't derecognize the asset. The lessor recognizes depreciation expense on the leased asset and rent revenue on the income statement.

Lease Accounting: Impact on Lessee Balance Sheet

A **lease liability** is recognized at the present value of the remaining lease payments

A **right-of-use asset** is recognized at an amount calculated as follows.

Amount of the lease obligation

- + Lease payments made to the lessor at or before the lease commencement date
 - Lease incentives received from the lessor
 - + Initial direct costs of right-of-use asset incurred by the lessee.
-

= Right-of-use asset

- The balance sheet presents **lease liabilities** and **right-of-use assets** *separately* (not the net amount)
- **Finance lease** assets are typically included in PPE, and lease liabilities are included with debt or other long term liabilities.
- **Operating lease** assets and liabilities are typically each reported in a separate line item, if material.

Lessor Accounting

- The lease classification tests for the lessor are identical to the tests used by the lessee to determine classification of a lease as a **financing** or **operating** lease.
- The lessor has, in substance, transferred control of the right-of-use asset and therefore has a **sales-type lease** if the lessee takes ownership or consumes a substantial portion of the underlying asset over the lease term.
- On the other hand, if the lease does not transfer control of the asset over the lease term, the lessor will generally use the **operating** approach in accounting for the lease.

Lessor Accounting for Sales-Type Leases

For a sales-type lease,

- the lessor accounts for the lease in a manner similar to the sale of an asset.
- the lessor generally records a long-term Lease Receivable and eliminates the leased asset.
- the lease receivable is computed at the present value of future lease payments received from the lessee.

A Comprehensive Illustration

Caterpillar Financial Services Corp. (a subsidiary of Caterpillar) and **Stole Construction** (Stole) sign a lease agreement dated January 1, 2025, that calls for Caterpillar to lease a backhoe to Stole beginning January 1, 2025. The terms and provisions of the lease agreement, and other pertinent data, are as follows.

- The term of the lease is five years. The lease agreement is non-cancelable, requiring equal rental payments of \$22,462.82 at the end of each year.
- The backhoe has a fair value at the commencement of the lease of \$100,000, an estimated economic life of five years with no residual value. Further, assume the backhoe has an \$85,000 cost to Caterpillar.
- Title transfers to Stole at the termination of the lease.
- Caterpillar sets the annual rental rate to earn a rate of return of 4 percent per year; Stole is aware of this rate.
- Assume the fiscal year ends on December 31 for both Caterpillar and Stole.

Lease Classification

First, note that the lease contract meets the criteria for classification as a **finance lease** because

- the present value of the lease payments is \$100,000 (derived next) and is equal to the fair value of the asset, and
- the lease term is five years which is equal to the economic life of the asset.
- title transfers to Stole at the end of the lease term.

That is, the terms of the contract are such that Stole (the lessee) will consume substantially the entire underlying asset over the lease term and therefore *control of the asset effectively transfers to Stole* at the inception of the lease contract. Similarly, Caterpillar (the lessor) classifies the lease as a **sales-type lease**.

Lessee Accounting: Finance Lease

- Stole computes the lease liability and the amount capitalized as a right-of-use asset as follows:

Payment	\$ 22,462.82
PV factor of Ordinary Annuity ($i = 4\%$, $n = 5$)	$\times \underline{4.4518}$
PV of lease payments	<u><u>\$ 100,000.00</u></u>

- Note that Stole uses Caterpillar's implicit interest rate of 4 percent to discount the cash flows. We assume that is known to Stole. Otherwise, Stole would use its incremental borrowing rate.

Lessee Accounting: Finance Lease

- Stole records the finance lease at its inception, i.e., on January 1, 2025, as:

Right-of-Use Asset	\$100,000
Lease Liability	\$100,000

Lessee Accounting: Finance Lease

Stole Construction: Lease Amortization Schedule

Date	Annual Lease Payment	Interest (4%) on Liability	Reduction of Lease Liability	Lease Liability
1/1/2025				\$100,000.00
31/12/25	\$ 22,462.82	\$ 4,000.00	\$18,462.82	81,537.18
31/12/26	22,462.82	3,261.49	19,201.33	62,335.85
31/12/27	22,462.82	2,493.43	19,969.39	42,366.46
31/12/28	22,462.82	1,694.66	20,768.16	21,598.30
31/12/29	22,462.82	<u>863.93</u>	<u>21,598.30</u>	0
	<u>\$112,314.10</u>	<u>\$12,314.10</u>	<u>\$100,000.00</u>	

- Stole records the first lease payment on December 31, 2025, as follows.

Interest Expense	\$ 4,000.00
Lease Liability	\$ 18,462.82
Cash	\$22,462.82

Lessee Accounting: Finance Lease

- To recognize amortization of the right-of-use asset at December 31, 2025 assuming Stole uses a straight-line depreciation policy:

Amortization Expense \$20,000

Right-of-Use Asset ($\$100,000 \div 5 \text{ years}$) \$20,000

- Stole records the second lease payment made on 31/12/2026 as follows:

Interest Expense	\$3,261.49
Lease Liability	\$19,201.33
Cash	\$22,462.82

Lessee Accounting: Finance Lease

- Stole's balance sheet as it relates to lease transactions at December 31, 2026.

<u>Noncurrent assets</u>	
Right-of-Use Asset (\$100,000– 40,000)	\$ 60,000.00
<u>Current liabilities</u>	
Lease liability	\$ 19,969.39
<u>Noncurrent liabilities</u>	
Lease liability	\$42,366.46

- Stole's fiscal 2026 income statement reports:

<u>Expenses</u>	
Interest expense	\$ 3,261.49
Amortization expense–Right-of-Use Asset	20,000.00

- What is the impact of the lease on Stole's SCF during fiscal 2026?

Lessor Accounting: Sales-Type Lease

- The journal entries on January 1, 2025, are as follows.

Lease Receivable	\$100,000
Sales Revenue	\$100,000
Cost of Goods Sold	\$85,000
Inventory	\$85,000

- On December 31, 2025, Caterpillar records receipt of the first lease payment:

Cash	\$22,462.82
Interest Revenue	\$4,000.00
Lease Receivable	18,462.82

- On December 31, 2026, Caterpillar records receipt of the second lease payment:

Cash	\$22,462.82
Interest Revenue	\$3,261.49
Lease Receivable	19,201.33

Lessor Accounting: Sales-Type Lease

- Caterpillar's balance sheet as it relates to lease transactions at December 31, 2025.

Current assets

Lease receivable	\$ 19,201.33
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Noncurrent assets (investments)

Lease receivable	\$62,335.85
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- Caterpillar's fiscal 2025 income statement reports:

Sales revenue	\$100,000.00
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– Cost of goods sold	85,000.00
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Gross Profit	\$ 15,000.00
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Interest revenue	4,000.00
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Operating Income	\$19,000.00
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Lessee Accounting for Operating Leases

- If a long-term lease **does not** meet any of the lease classification tests for a finance lease, a lessee should classify it as an *operating lease*.
- For leases classified as operating, the lessee records
 - a Right-of-Use asset and lease liability at commencement of the lease, **similar** to the finance lease approach.
 - However, **unlike** a finance lease, the lessee records the same amount for lease expense each period over the lease term.

Let's modify the previous example...

Caterpillar Financial Services Corp. (a subsidiary of Caterpillar) and **Stole Construction** (Stole) sign a lease agreement dated January 1, 2025, that calls for Caterpillar to lease a backhoe to Stole beginning January 1, 2025. The terms and provisions of the lease agreement, and other pertinent data, are as follows.

- The lease agreement is non-cancelable with a term of 3 years requiring equal rental payments of \$22,446.69 at the end of each year.
 - The backhoe has a cost and fair value at the commencement of the lease of \$100,000. The estimated economic life of the backhoe is five years with a residual value of \$20,000.
 - There are no renewal options and the backhoe is not of a specialized nature so that it reverts back to Caterpillar at the termination of the lease.
 - Stole doesn't know Caterpillar's implicit rate of return and therefore uses its incremental borrowing rate is 6% to evaluate the lease.
- Note that, being a long-term lease, the lease would still be treated as a finance lease under IFRS. *However, under US GAAP, we have to further determine whether the above lease qualifies as an operating lease vs. a finance lease.*

Lessee Accounting: Operating Lease

Fair value of leased equipment	\$100,000.00
Present Value of three end-of-year lease payments at an interest rate of 6% ($\$22,446.69 \times 2.6730$ ($PVANN_{3,6\%}$))	<u>\$60,000.00</u>

Test	Assessment
1. Transfer of ownership test	Transfer of ownership does not occur; the asset reverts to Caterpillar at the end of the lease.
2. Purchase option test	There is no purchase option in the lease.
3. Lease term test	The lease term is 60 percent ($3 \div 5$) of the economic life of the asset, which is less than a major part of the life of the asset.
4. Present value test	From above, the present value of the lease payments is \$60,000, which is 60% ($\$60,000 \div \$100,000$) of the fair value of the backhoe. Therefore, the lease fails the present value test.
5. Alternative use test	As indicated, the equipment is not of a specialized nature and is expected to be of use to Caterpillar at the end of the lease.

- Stole should classify lease as operating as none of the finance lease tests are met.

Lessee Accounting: Operating Lease

- Stole makes the following entry on January 1, 2025 to record this operating lease

Right-of-Use Asset \$60,000.00

Lease Liability \$60,000.00

Stole Construction

Lease Liability Amortization Schedule

Date	Annual Lease Payment	Interest (6%) on Liability	Reduction of Lease Liability	Lease Liability
1/1/2025				\$60,000.00
31/12/25	\$22,446.69	\$ 3,600.00	\$18,846.69	41,153.31
31/12/26	22,446.69	2,469.20	19,977.49	21,175.82
31/12/27	<u>22,446.69</u>	<u>1,270.87</u>	<u>21,175.82</u>	0.00
	<u>\$67,340.07</u>	<u>7,340.07</u>	<u>\$60,000.00</u>	

Lessee Accounting: Operating Lease

Lease Expense Schedule

Date	(A) Lease Expense (Straight-Line)	(B) Interest (6%) on Liability	(C) Amortization of ROU Asset <i>(derived)</i> (A – B)	(D) Carrying Value of ROU Asset (D – C)
1/1/2025				\$60,000.00
31/12/2025	\$22,446.69	\$ 3,600.00	\$18,846.69	41,253.31
31/12/2026	22,446.69	2,469.20	19,977.49	21,175.82
31/12/2027	<u>22,446.69</u>	<u>1,270.87</u>	<u>21,175.82</u>	0.00
	<u>\$67,340.07</u>	<u>7,340.07</u>	<u>\$60,000.00</u>	

- Stole makes the following entries during fiscal 2025 on December 31, 2025:

Lease Expense	\$22,446.69		
	Cash	\$22,446.69	
Lease Liability	\$18,846.69		
	Right-of-Use Asset	\$18,846.69	

Lessee Accounting: Operating Lease

- Stole makes the following entries on December 31, 2026, the second year of the lease:

Lease Expense	\$22,446.69		
	Cash	\$22,446.69	
Lease Liability	\$19,977.49		
	Right-of-Use Asset	\$19,977.49	

- Stole makes the following entries on December 31, 2027, the third and final year of the lease.

Lease Expense	\$22,446.69		
	Cash	\$22,446.69	
Lease Liability	\$21,175.82		
	Right-of-Use Asset	\$21,175.82	

Lessor Accounting: Operating Lease

- Applying the same classification tests used by Stole, Caterpillar classifies the lease as an operating lease because **none** of the finance lease tests are met.
- Under the operating lease method, Caterpillar (the lessor) continues to recognize the asset on its balance sheet and recognizes lease revenue (generally on a straight-line basis) in each period.
- Caterpillar depreciates the leased asset using its depreciation policy on such assets. We will assume it is straight-line over 5 years.

Lessor Accounting: Operating Lease

- To illustrate the operating lease method, Caterpillar records the lease payment on a straight-line basis on December 31, 2025, 2026, and 2027, as follows.

Cash	22,446.69	
	Lease Revenue	22,446.69

- Caterpillar also records depreciation expense on the leased equipment given a cost basis of \$100,000, a residual value of \$20,000, and a five-year economic life, as follows.

Depreciation Expense	16,000.00	
	Accumulated Depreciation—Equipment	16,000.00

Impact of Operating and Finance leases on the SCF

Unlike IFRS, under US GAAP, the statement of cash flows will be impacted by the lease classification in a similar manner to the income statement:

- **Operating lease:** Cash flow from operating activities includes the *entire lease payment*.
- **Finance lease:** The lease payments include cash payment of accrued interest and reduction of the principal balance of the lease liability:
 - The *interest portion* is subtracted on the income statement and, therefore reduces net cash flows from operating activities (CFFO).
 - The *principal portion* is considered a financing activity and therefore reduces cash flows from financing activities (CFFF).
 - All else equal, net cash flows from operating activities will therefore be *higher* for finance leases (relative to operating leases) by the amount of the payment allocated to reduction of the lease liability.

Microsoft's Lease Disclosures

- On its fiscal 2024 balance sheet, Microsoft reports assets relating to both operating and finance leases

(In millions)

June 30,	2024	2023
Assets		
Current assets:		
Cash and cash equivalents	\$ 18,315	\$ 34,704
Short-term investments	57,228	76,558
Total cash, cash equivalents, and short-term investments	75,543	111,262
Accounts receivable, net of allowance for doubtful accounts of \$830 and \$650	56,924	48,688
Inventories	1,246	2,500
Other current assets	26,021	21,807
Total current assets	159,734	184,257
Property and equipment, net of accumulated depreciation of \$76,421 and \$68,251	135,591	95,641
Operating lease right-of-use assets	18,961	14,346
Equity and other investments	14,600	9,879
Goodwill	119,220	67,886
Intangible assets, net	27,597	9,366
Other long-term assets	36,460	30,601
Total assets	\$ 512,163	\$ 411,976

Finance Lease ROU Assets are included in PP&E

Microsoft's Lease Disclosures

- Microsoft reports the lease liabilities on its fiscal 2024 balance sheet

Current liabilities:

Accounts payable	\$ 21,996	\$ 18,095
Short-term debt	6,693	0
Current portion of long-term debt	2,249	5,247
Accrued compensation	12,564	11,009
Short-term income taxes	5,017	4,152
Short-term unearned revenue	57,582	50,901
Other current liabilities	19,185	14,745
Total current liabilities	125,286	104,149
Long-term debt	42,688	41,990
Long-term income taxes	27,931	25,560
Long-term unearned revenue	2,602	2,912
Deferred income taxes	2,618	433
Operating lease liabilities	15,497	12,728
Other long-term liabilities	27,064	17,981
Total liabilities	243,686	205,753

Includes current portions of both operating and finance leases

Includes non-current portion of finance leases

Microsoft's Lease Disclosures

We have operating and finance leases for datacenters, corporate offices, research and development facilities, Microsoft Experience Centers, and certain equipment. Our leases have remaining lease terms of less than 1 year to 17 years, some of which include options to extend the leases for up to 5 years, and some of which include options to terminate the leases within 1 year.

The components of lease expense were as follows:

(In millions)

Year Ended June 30,	2024	2023	2022
Operating lease cost	\$ 3,555	\$ 2,875	\$ 2,461
Finance lease cost:			
Amortization of right-of-use assets	\$ 1,800	\$ 1,352	\$ 980
Interest on lease liabilities	734	501	429
Total finance lease cost	\$ 2,534	\$ 1,853	\$ 1,409

Supplemental cash flow information related to leases was as follows:

(In millions)

Year Ended June 30,	2024	2023	2022
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash flows from operating leases	\$ 3,550	\$ 2,706	\$ 2,368
Operating cash flows from finance leases	734	501	429
Financing cash flows from finance leases	1,286	1,056	896
Right-of-use assets obtained in exchange for lease obligations:			
Operating leases	6,703	3,514	5,268
Finance leases	11,633	3,128	4,234

Microsoft's Lease Disclosures

Supplemental balance sheet information related to leases was as follows:

(In millions, except lease term and discount rate)

June 30,	2024	2023
Operating Leases		
Operating lease right-of-use assets	\$ 18,961	\$ 14,346
Other current liabilities	\$ 3,580	\$ 2,409
Operating lease liabilities	15,497	12,728
Total operating lease liabilities	\$ 19,077	\$ 15,137
Finance Leases		
Property and equipment, at cost	\$ 32,248	\$ 20,538
Accumulated depreciation	(6,386)	(4,647)
Property and equipment, net	\$ 25,862	\$ 15,891
Other current liabilities	\$ 2,349	\$ 1,197
Other long-term liabilities	24,796	15,870
Total finance lease liabilities	\$ 27,145	\$ 17,067
Weighted Average Remaining Lease Term		
Operating leases	7 years	8 years
Finance leases	12 years	11 years
Weighted Average Discount Rate		
Operating leases	3.3%	2.9%
Finance leases	3.9%	3.4%

Microsoft's Lease Disclosures

- Footnotes disclose a **schedule of future lease payments**

The following table outlines maturities of our lease liabilities as of June 30, 2024:

(In millions)

Year Ending June 30,		Operating Leases		Finance Leases
2025	\$	4,124	\$	3,311
2026		3,549		3,021
2027		2,981		3,037
2028		2,405		3,026
2029		1,924		2,638
Thereafter		6,587		19,116
Total lease payments		21,570		34,149
Less imputed interest		(2,493)		(7,004)
Total	\$	19,077	\$	27,145

As of June 30, 2024, we had additional operating and finance leases, primarily for datacenters, that had not yet commenced of \$8.6 billion and \$108.4 billion, respectively. These operating and finance leases will commence between fiscal year 2025 and fiscal year 2030 with lease terms of 1 year to 20 years.

Using the above schedule, estimate the weighted average discount rate on Microsoft's outstanding operating and finance leases at June 30, 2024.

Microsoft's Lease Disclosures

- *Suppose the amortization expense on the ROU assets for finance leases during fiscal 2025 would be \$1,900M. What would be the overall impact of Microsoft's operating and finance leases on its income before taxes and its statement of cash flows during fiscal 2025?*
- *The carrying value of Microsoft's finance lease liabilities at June 30, 2024 is \$27,145M. Using the information provided, estimate (1) the carrying value of the finance lease liabilities at June 30, 2025, and (2) the current and non-current portions of the finance lease liabilities at June 30, 2025.*

Summary of Current Leases Standard

	Operating Lease	Finance Lease
Balance Sheet (same for both operating and finance leases)	• All leases are recognized on the balance sheet (except leases with a term of less than 12 months). • Lease asset is reported as either PPE or a "right-of-use" asset that is amortized over the lease life. • Lease liability is reduced by principal payments each period, like a mortgage. • Accounting treatment is similar to recording a PPE asset that is purchased and financed with borrowed money (both the asset and liability are reported on the balance sheet).	
Income Statement	• Rent expense is recognized for the straight-line amortization of the total lease payments plus up-front costs.	• Straight-line amortization expense of the right-of-use asset, <i>plus</i> • Interest expense is recognized on the lease liability.
Statement of Cash Flows	• Lease payments are classified as operating cash flow.	• Interest portion of lease payments is classified as operating cash flow. • Principal portion of lease payments is classified as financing cash flow.